Module - 2

Properties of the Integers

Well ordering principle: Every nonempty subset of Z^+ contains a least element.

Mathematical Induction: The method of mathematical induction is based on a principle called the Induction principle.

Induction principle:

Let $S(n)$ denote an open statment that involves a positive integer n.

(1) $S(1)$ is true

(2) If whenever $S(K)$ is true for some particular, but arbitrarily chosen $K \in Z^+$, the $S(K+1)$ is true Then $S(n)$ is true for all $n \in Z^+$.

Method of Mathematical Induction:

(i) Basic step: verify that the statement $S(1)$ is true, , i.e, verify that $S(n)$ is true for $n=1$

(ii) Induction step: Assuming that $S(K)$ is true, where K is an integer ≥ 1, S.T. $S(K+1)$ is true.

Examples:

① Prove that by mathematical Induction.

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$$

Sol:

$$S(n) = 1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$$

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Basic step: $n=1$

$$1^2 = \frac{1(2-1)(2+1)}{3} = \frac{(1)(3)}{3} = 1$$

$S(1)$ is true,

Induction step; Assume that $S(n)$ is true for $n=k$ where $k \geq 1$. Then

$$S(k) = 1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 = \frac{k(2k-1)(2k+1)}{3}$$

adding $(2k+1)^2$ on both sides, we obtain,

$$1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 + (2k+1)^2 = \frac{k(2k-1)(2k+1)}{3} + (2k+1)^2$$

$$= \frac{2k+1}{3}\left[k(2k-1) + 3(2k+1)\right]$$

$$= \frac{2k+1}{3}\left[2k^2 + 5k + 3\right]$$

$2k^2 + 2k + 3k + 3$

$2k(k+1) + 3(k+1)$

$$= \frac{2k+1}{3}(k+1)(2k+3)$$

$$= \frac{(k+1)(2k+1)(2k+3)}{3}$$

This is Exactly statement of $S(k+1)$

Thus the statement $S(k+1)$ is true whenever the statement $S(k)$, where $k \geq 1$ is true,

Hence by mathematical Induction, it follows that $S(n)$ is true for all Integers $n \geq 1$.

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(2) Prove by mathematical Induction.

$$1\times 3 + 2\times 4 + 3\times 5 + \cdots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$$

Sol: $S(n) = \cancel{1^2 + 3^2 + \cdots + (2n-1)^2}$

Basic step: $n=1$

$$1\times 3 = \frac{(1)(1+1)(2+7)}{6} = \frac{(2)(9)}{6} = 3$$

$S(1)$ is true,

Induction step: Assuming that $S(n)$ is true for $n=k$ where $k\geq 1$, then

$$S(k) = 1\times 3 + 2\times 4 + 3\times 5 + \cdots + k(k+2) = \frac{k(k+1)(2k+7)}{6}$$

Adding $(k+1)(k+3)$ on B.S

$$1\times 3 + 2\times 4 + \cdots + k(k+2) + (k+1)(k+3) = \frac{k(k+1)(2k+7)}{6} + (k+1)(k+3)$$

$$= \frac{(k+1)}{6}\left[k(2k+7) + 6(k+3)\right]$$

$$= \frac{(k+1)}{6}\left[2k^2 + 13k + 18\right]$$

$$= \frac{(k+1)}{6}\left[2k^2 + 9k + 4k + 18\right]$$

$$= \frac{(k+1)}{6}\left[k(2k+9) + 2(2k+9)\right]$$

$$= \frac{(k+1)}{6}\left[(k+2)(2k+9)\right]$$

This is precisely the statement $S(k+1)$

Thus, the statement $S(k+1)$ is true whenever the statement $S(k)$ is true for $k\geq 1$

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③ Prove that for each $n \in \mathbb{Z}^+$,

$$1^2+2^2+3^2+\cdots+n^2=\frac{1}{6}n(n+1)(2n+1)$$

Sol: $S(n)=1^2+2^2+\cdots+n^2=\frac{1}{6}n(n+1)(2n+1)$

Basic step: $n=1$

$$\Rightarrow 1^2\cdot 1=\frac{1}{6}(1)(2)(3)=1 \Rightarrow S(1) \text{ is true}$$

Induction step: Assume that $S(n)$ is true for $n=k$, $k\geq 1$

$$S(k)=1^2+2^2+\cdots+k^2=\frac{1}{6}k(k+1)(2k+1)$$

Adding $(k+1)^2$ on B.S we get

$$1^2+2^2+\cdots+k^2+(k+1)^2=\frac{1}{6}k(k+1)(2k+1)+(k+1)^2$$

$$=\frac{(k+1)}{6}\{k(2k+1)+6k+6\}$$

$$=\frac{k+1}{6}[2k^2+k+6k+6]$$

$$=\frac{k+1}{6}[2k^2+7k+6]$$

$$=\frac{k+1}{6}(2k+3)(k+2)$$

$$=\frac{(k+1)(k+2)(2k+3)}{6}$$

This is precisely the statement $S(k+1)$.

Thus the statement $S(k+1)$ is true whenever the statement $S(k)$ is true for $k\geq 1$.

This is the required result by method of Induction.

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(4) If n is any +ve integer, prove that

$$1\cdot 2 + 2\cdot 3 + 3\cdot 4 + \cdots + n(n+1) = \frac{1}{3} n(n+1)(n+2)$$

Sol: $n=1$ $\Rightarrow 1\cdot 2 = \frac{1}{3}(1)(2)(3) = 2 \Rightarrow S(1)$ is true.

Assume that $S(n)$ is true for $n=k$ where $k \geq 1$

$S(k)$: $1\cdot 2 + 2\cdot 3 + \cdots + k(k+1) = \frac{1}{3} k(k+1)(k+2)$

adding $(k+1)(k+2)$ on B.S we get

$$1\cdot 2 + 2\cdot 3 + \cdots + k(k+1) + (k+1)(k+2) = \frac{1}{3} k(k+1)(k+2) + (k+1)(k+2)$$

$$= \frac{(k+1)(k+2)}{3}[k+3]$$

$$= \frac{(k+1)(k+2)(k+3)}{3}$$

This is exactly the statement $S(k+1)$, Thus the Statement $S(k+1)$ is true.

$\Rightarrow S(k)$ is true for $k \geq 1$.

(5) prove ~~that~~ by mathematical induction that, for every positive integer n, 5 divides $n^5 - n$.

Sol: $S(n)$: 5 divides $n^5 - n$.

$n=1 \Rightarrow$ 5 divides $1^5 - 1 = 0$ This statement is true.

Assuming $S(n)$ is true for $n=k$, $k \geq 1$

5 divides $k^5 - k$

This means that $k^5 - k$ is a multiple of 5

i.e, $k^5 - k = 5m$, for some +ve integer m,

consequently we find that

$$\cancel{(k+1)^5} \quad (k+1)^5 - (k+1) = (k^5 + 5k^4 + 10k^2 + 5k + 1) - (k+1)$$

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$\because$ Using Binomial thm $(x+y)^n = \sum_{r=0}^{n} {}^nC_r x^r y^{n-r}$

$$(k+1)^5 = \sum_{r=0}^{5} {}^5C_r k^r 1^{5-r}$$

$$= {}^5C_0 + {}^5C_1 k^1 + {}^5C_2 k^2 + {}^5C_3 k^3 + {}^5C_4 k^4 + {}^5C_5 k^5$$

$$= 1 + 5k + 10k^2 + 10k^3 + 5k^4 + k^5$$

$$\therefore (k+1)^5 - (k+1) = (k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + \cancel{1}) - (k + \cancel{1})$$

$$= (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k) \quad \because m = k^5 - k$$

$$= 5m + 5(k^4 + 2k^3 + 2k^2 + k)$$

$$= 5[m + k^4 + 2k^3 + 2k^2 + k]$$

This shows that $(k+1)^5 - (k+1)$ is a multiple of 5, i.e, 5 divides $(k+1)^5 - (k+1)$, This is precisely the statement of $S(k+1) \Rightarrow S(k+1)$ is true $\Rightarrow S(k)$ is true $k \geq 1$

(6) By M.I Prove that for every +ve integer n, the number $A_n = 5^n + 2 \cdot 3^{n-1} + 1$ is a multiple of 8.

Sol) $A_1 = 5^1 + 2 \cdot 3^0 + 1 = 8$

the number A_n is a multiple of 8 for $n = 1$.

Using the definition A_n,

$$A_{k+1} - A_k = (5^{(k+1)} + 2 \cdot 3^k + \cancel{1}) - (5^k + 2 \cdot 3^{k-1} + \cancel{1})$$

$$= (5-1)5^k + 2(3-1)3^{k-1} = 4 \cdot 5^k + 4 \cdot 3^{k-1} = 4(5^k + 3^{k-1})$$

Since 5 & 3 are odd, 5^k & 3^{k-1} are also odd.

$\therefore 5^k + 3^{k-1}$ is even, Hence $4(5^k + 3^{k-1})$ is multiple of 8

i.e, $A_{k+1} - A_k$ is a multiple of 8

because A_k is a multiple of 8, A_{k+1} is also multiple of 8

A_k is also multiple of 8

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(7) P.T $4n < (n^2-7)$ for all positive integers $n \geq 6$

Sol Here, we have to p. the statement that

$S(n)$: $4n < (n^2-7)$ is true for all +ve integers $n \geq n_0$, where $n_0 = 6$.

$S(6)$: $4 \times 6 < (6^2-7)$

$24 < 29$ is true.

Thus, $S(n)$ is true for $n = n_0 = 6$.

Assume that $S(n)$ is true for $n = k$ where $k \geq 6$

$4k < (k^2-7)$ for $k \geq 6$

$$\begin{aligned} 4(k+1) &= 4k+4 \\ &< (k^2-7)+4 \\ &< (k^2-7)+(2k+1) \quad \because k \geq 6, \; 2k+1 \geq 13 > 4 \\ &= (k+1)^2-7 \end{aligned}$$

This shows that $S(k+1)$ is true

$S(n)$ is true for all +ve integers $n \geq 6$ by Mathematical Induction.

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*) prove that every positive integer $n \geq 24$ can be written as a sum of 5's and/or 7's.

Soln:- $S(n)$: n can be written as a sum of 5's and/or 7's is true for all integers $n \geq 24$.

Basic step:- Consider $S(24)$

$$24 = (4+4) + (5+5)$$

This shows that $S(24)$ is true.

Induction step

we assume that $S(n)$ is true for $n = k$, where $k \geq 24$

ie $k = (7+7+ - -) + (5+5+ - -)$

Suppose this representation of k has r no of 7 & s no of 5

Since $k \geq 24$ $r \geq 2$, $s \geq 2$.

Now $k+1 = \left\{ \underbrace{[7+7+ - - -]}_{r} + \underbrace{[5+5+ - - -]}_{s} \right\} + 1$

$= \underbrace{[7+7+ - -]}_{(r-2)} + (7+7) + \underbrace{[5+5+ - - - -]}_{s} + 1$

$(r-2) \leftarrow = [7+7+ - -] + [5+5+ - -] \rightarrow (s+3)$

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This shows that $(k+1)$ is a sum of 7's & 5's

Thus $S(k+1)$ is true.

Hence by Mathematical Induction

$S(n)$ is true for all positive integers $n \geq 24$

(*) By mathematical Induction. prove that

$(n!) \geq 2^{n-1}$ for all integers $n \geq 1$.

<u>Soln</u>:- $S(n)$: $n! \geq 2^{n-1}$ $\forall n \geq 1$

<u>Basic Step</u>: For $n=1$

$S(1)$: $1! \geq 2^{1-1}$

$1! \geq 1$

$1 \geq 1$.

$\therefore$ $S(1)$ is true.

<u>Induction Step</u>:-

we assume that $S(n)$ is true for $n=K$ where $K \geq 1$

ie $K! \geq 2^{K-1}$

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$$2^k = 2 \cdot 2^{k-1}$$
$$\leq 2 \cdot k!$$
$$< (k+1) \cdot k! \qquad (\because 2 < k+1 \text{ for } k \geq 1)$$
$$= (k+1)!$$

$$(k+1)! \geq 2^k$$

This is nothing but $S(k+1)$ is true.

whenever $S(k)$ is true for $k \geq 1$

Thus by Mathematical induction

$S(n)$ is true for all integers $n \geq 1$

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(*) Let $a_0 = 1$, $a_1 = 2$, $a_2 = 3$ & $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$. P.T $\underline{a_n \leq 3^n}$ $\forall$ integers n

<u>Soln</u> :- Consider $S(n)$: $a_n \leq 3^n$

<u>Basic step</u>

For

$n = 0$ $\quad$ $n = 1$ $\quad$ $n = 2$

$a_0 = 1 \leq 3^0$ $\quad$ $a_1 = 2 \leq 3^1$ $\quad$ $a_2 = 3 \leq 3^2$

Thus, $S(n)$ is true for $n = 0, 1, 2$.

<u>Induction step</u>

Now assume that $S(n)$ is true for $n = 0, 1, 2, \ldots k$, where $k \geq 2$. ie $a_k \leq 3^k$ $\forall$

Then we note that

$$a_{k+1} = a_k + a_{k-1} + a_{k-2} \quad (\because \text{by defn of } a_n)$$

$$\leq 3^k + 3^{k-1} + 3^{k-2}$$

$$\leq 3^k + 3^k + 3^k$$

$$= 3 \times 3^k$$

$$= 3^{k+1}$$

ie $a_{k+1} \leq 3^{k+1}$

Thus $S(k+1)$ is true, whenever $S(k)$ is true.

Hence by Mathematical Induction, $S(n)$ is true for all positive integers 'n'.

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Recursive Definition / Recursive Rule.

A sequence is an arrangement of elements of a set in order as the first element, the 2nd element & so on. There are two methods to describing a sequences.

(1) The Explicit method: The general term of the sequences is explicitely indecated.

(2) Recursive method: In this method, first few terms of the sequences are explicitly indecated and the general term is specified through a formula which indicates how to obtain new terms of the sequence from the known terms.

Examples:

① A sequence $\{a_n\}$ is defined recursively by $a_1 = 4$, $a_n = a_{n-1} + n$ for $n \geq 2$. Find a_n in explicit form.

Sol: Recursive formula $a_n = a_{n-1} + n$

$$\Rightarrow a_n = [a_{n-2} + (n-1)] + n$$
$$= [a_{n-3} + (n-2)] + (n-1) + n$$
$$= [a_{n-4} + (n-3)] + (n-2) + (n-1) + n$$
$$\vdots$$
$$= a_1 + 2 + 3 + \cdots + n$$
$$= (a_1 - 1) + (1 + 2 + 3 + \cdots + n) \longrightarrow ①$$ by adding & sub. 1

Using $a_1 = 4$ & $1 + 2 + 3 + \cdots + n = \frac{1}{2} n(n+1)$

$$① \Rightarrow \boxed{a_n = 3 + \frac{1}{2} n(n+1)}$$

This is explicit formula for a_n.

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(2) Find an explicit definition of the sequence defined recursively by $a_1 = 7$, $a_n = 2a_{n-1} + 1$ for $n \geq 2$

Sol) Using Recursive definition

$$a_n = 2a_{n-1} + 1$$

$$= 2[2a_{n-2} + 1] + 1$$

$$= 2[2\{2a_{n-3} + 1\} + 1] + 1 = 2^3 a_{n-3} + 2^2 + 2 + 1$$

$$\vdots$$

$$= 2^{n-1} a_{n-(n-1)} + 2^{n-2} + 2^{n-3} + \dots + 2^2 + 2 + 1$$

$$= 2^{n-1} a_1 + (1 + 2 + 2^2 + \dots + 2^{n-3} + 2^{n-2}) \longrightarrow (1)$$

Using $a_1 = 7$ &

$$1 + a + a^2 + \dots + a^{n-1} = \frac{a^n - 1}{a - 1} \text{ for } a > 1.$$

$$(1) \Rightarrow a_n = 7 \times 2^{n-1} + (2^{n-1} - 1)$$

$$= (7 + 1) 2^{n-1} - 1$$

$$= (8 \times 2^{n-1}) - 1$$

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3) A sequence $\{C_n\}$ is defined recursively by $C_n = 3C_{n-1} - 2C_{n-2}$ for all $n \geq 3$, with $C_1 = 5$ & $C_2 = 3$ as the initial conditions, S.T $C_n = -2^n + 7$.

Sol given for $n = 3$

$$C_n = 3C_{n-1} - 2C_{n-2}$$

$$C_3 = 3C_2 - 2C_1 = 3\times3 - 2\times5 = 9 - 10 = -1$$

$$= -2^3 + 7$$

This expression shows that $C_n = -2^n + 7 \longrightarrow (1)$ is true for $n=3$. Assume that the Eqn (1) is true for $n = 3, 4, \ldots k$.

Using Recursive Relation, we find that

$$C_{k+1} = 3C_{(k+1)-1} - 2C_{(k+1)-2}$$

$$= 3C_k - 2C_{k-1}$$

$$= 3\times(-2^k + 7) - 2(-2^{k-1} + 7)$$

$$= (-3\times2^k + 21) + (2^k - 14)$$

$$= (-3+1)2^k + (21-14)$$

$$= -2^{k+1} + 7.$$

This show that the expression

$C_n = -2^n + 7$ holds for $n = k+1$.

The expression holds for all integers $n \geq 3$ using mathematical induction.

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(6) For the Fibonacci sequence $F_0, F_1, F_2, \ldots$, P.T.

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

→ For $\boxed{n=0}$

$$F_0 = \frac{1}{\sqrt{5}}\left[\underbrace{\left(\frac{1+\sqrt{5}}{2}\right)^0}_{\Rightarrow 1} - \underbrace{\left(\frac{1-\sqrt{5}}{2}\right)^0}_{1}\right] = 0.$$

For $\boxed{n=1}$

$$F_1 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^1 - \left(\frac{1-\sqrt{5}}{2}\right)^1\right]$$

$$= \frac{1}{\sqrt{5}}\left[\frac{1}{2} + \frac{\sqrt{5}}{2} - \frac{1}{2} + \frac{\sqrt{5}}{2}\right]$$

$$= \frac{1}{\sqrt{5}}\left(2\,\frac{\sqrt{5}}{2}\right)$$

$$\boxed{F_1 = 1}$$

Which are true for $n=0$ and $n=1$.
Assume that the result is true for $n = 0, 1, 2, \ldots, k$, where $k \geq 1$. Then we find that
Note: $F_n = F_{n-1} + F_{n-2}$

$$F_{k+1} = F_k + F_{k-1}$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right] + \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k-1}\left\{\frac{1+\sqrt{5}}{2}+1\right\} - \left(\frac{1-\sqrt{5}}{2}\right)^{k-1}\left\{\frac{1+\sqrt{5}}{2}+1\right\}\right]$$

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$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{K-1}\left\{\frac{2}{2}\left(\frac{1+\sqrt{5}+2}{2}\right)\right\} - \left(\frac{1-\sqrt{5}}{2}\right)^{K-1}\left\{\frac{2}{2}\left(\frac{1+\sqrt{5}}{2}+1\right)\right\}\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{K-1}\left(\frac{6+2\sqrt{5}}{4}\right) - \left(\frac{1-\sqrt{5}}{2}\right)^{K-1}\left(\frac{6-2\sqrt{5}}{4}\right)\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{K-1}\left(\frac{1+\sqrt{5}}{2}\right)^{2} - \left(\frac{1-\sqrt{5}}{2}\right)^{K-1}\left(\frac{1-\sqrt{5}}{2}\right)^{2}\right]$$

$$= \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{K+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{K+1}\right]$$

This shows that the required result is true for $n = K+1$. Hence by mathematical induction, the result is true for all non-negative integers n.

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